Santiago G. Medina-Muñoz

PhD Candidate in Integrative Biology (defense expected Fall 2026) — Population Genetics & Computational Biology

Irapuato, Guanajuato, México

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I study how demographic history—migration, admixture, and drift—has shaped genetic variation in human populations, and what that variation means for health. My doctoral work combines 500 whole genomes from across Mexico with coalescent-based demographic inference to reconstruct the pre-Columbian and post-contact history of Indigenous and admixed Mexican lineages. I have also contributed to the Mexican Biobank, characterizing ancestry and clinically relevant variation in Hispanic populations. I am now moving into long-read genomics and pangenomes, starting with ten pilot genome assemblies. Earlier, at the Stowers Institute, I studied how codon composition shapes mRNA stability and translation.

Education

PhD in Integrative Biology
CINVESTAV, LANGEBIO — Irapuato, Mexico — 2022–Present, Defense expected Fall 2026

Topic: Evolutionary history of pre-Columbian Mexico. Advisors: Andrés Moreno-Estrada & Aaron Ragsdale.

MSc in Integrative Biology
CINVESTAV, LANGEBIO — Irapuato, Mexico — 2020–2022

Topic: Demographic inference in admixed populations. Advisor: Andrés Moreno-Estrada.

BSc in Agro-genomic Sciences
Universidad Nacional Autónoma de México — León, Mexico — 2013–2017

Publications

Preprints

Snodgrass S., Li F., Mambakkam S., **Medina-Muñoz S.G.**, . . . , Ross-Ibarra J. Maize genetic diversity is largely unstructured by human ethnolinguistic diversity in its center of origin. *bioRxiv*, 2026. doi:10.1101/2026.02.02.703385

Under review

Medina-Muñoz S.G., et al. The genomic history of pre-colonial contacts across the Americas. Under review at *Nature*.

Peer-reviewed journal articles

8. Barberena-Jonas C., **Medina-Muñoz S.G.**, et al. Clinical genetic variation across Hispanic populations in the Mexican Biobank. *Nature Medicine*, 32:725–735, 2026. doi:10.1038/s41591-025-04100-z

7. Sohail M., Palma-Martínez M.J., . . . , **Medina-Muñoz S.G.**, et al. Mexican Biobank advances population and medical genomics of diverse ancestries. *Nature*, 622(7984):775–783, 2023. doi:10.1038/s41586-023-06560-0

6. **Medina-Muñoz S.G.**, Ortega-Del Vecchyo D., Cruz-Hervert L.P., Ferreyra-Reyes L., García-García L., Moreno-Estrada A., Ragsdale A.P. Demographic modeling of admixed Latin American populations from whole genomes. *The American Journal of Human Genetics*, 110(10):1804–1816, 2023. doi:10.1016/j.ajhg.2023.08.015

5. *Diez M., ***Medina-Muñoz S.G.**, Castellano L.A., et al. iCodon customizes gene expression based on the codon composition. *Scientific Reports*, 12(1):12126, 2022. doi:10.1038/s41598-022-15526-7

4. Rodríguez-Rodríguez J.E., Ioannidis A.G., **Medina-Muñoz S.G.**, Barberena-Jonas C., Blanco-Portillo J., Quinto-Cortés C.D., Moreno-Estrada A. The genetic legacy of the Manila galleon trade in Mexico. *Philosophical Transactions of the Royal Society B*, 377(1852):20200419, 2022. doi:10.1098/rstb.2020.0419

3. Jiménez-Kaufmann A., Chong A.Y., . . . , **Medina-Muñoz S.G.**, et al. Imputation performance in Latin American populations: improving rare variant representation with the inclusion of Native American genomes. *Frontiers in Genetics*, 12:719791, 2022. doi:10.3389/fgene.2021.719791

2. **Medina-Muñoz S.G.**, Kushawah G., Castellano L.A., et al. Crosstalk between codon optimality and cis-regulatory elements dictates mRNA stability. *Genome Biology*, 22:1–23, 2021. doi:10.1186/s13059-020-02251-5

1. Wu Q., **Medina-Muñoz S.G.**, Kushawah G., et al. Translation affects mRNA stability in a codon-dependent manner in human cells. *eLife*, 8:e45396, 2019. doi:10.7554/eLife.45396

** equal contribution. 832 citations, h-index 8 (Google Scholar, October 2026). Full list on Google Scholar.*

Research Experience

PhD Fellow — Computational Genomics & Population Genetics
CINVESTAV LANGEBIO, Irapuato, Mexico
2020–Present
Sup. A. Moreno-Estrada & A. Ragsdale

Reconstructing fine-scale demographic history of Indigenous and admixed Mexican lineages from whole-genome sequence data using coalescent and site-frequency-spectrum methods.

Honorary Fellow
University of Wisconsin–Madison, USA
Summer 2023
Sup. A. Ragsdale

Extended PhD project on demographic inference of Mexican populations using whole-genome data.

Bioinformatician
Stowers Institute, Kansas City, USA
2017–2020
PI A. Bazzini

Studied how gene regulatory information is encoded in the coding sequence through codon optimality.

Undergraduate Research Thesis
Stowers Institute, Kansas City, USA
2016–2017
PI A. Bazzini

Thesis: “The regulatory grammar of codon optimality.”

Stowers Summer Scholar
Stowers Institute, Kansas City, USA
Summer 2015
PI J. Zeitlinger

Developed a pipeline for analyzing protein-DNA binding interactions with ChIP-nexus data.

Undergraduate Summer Research

CINVESTAV LANGEBIO, Irapuato, Mexico

Summer 2014
PI C. Abreu-Goodgear

Investigated conservation of micro-RNA targets across vertebrate species using RNA-seq data.

Technical Skills

Population genetics: demographic inference (moments, msprime, ABC), coalescent simulation, site-frequency-spectrum modeling, admixture & local ancestry inference.

Bioinformatics: whole-genome variant calling (GATK, bcftools), read alignment, RNA-seq & ChIP-nexus analysis pipelines.

Programming & tools: Python, R, Bash, Snakemake, Git, HPC/Slurm cluster computing.

Funding, Fellowships & Awards

Graduate

• Travel Grant (€1,000) — *EMBO | EMBL Symposium: Reconstructing the Human Past, EMBL Heidelberg* (Sept 2026)
• EMBO | EMBL Symposia Fellowship — *fully funded travel & participation, EMBO Symposium: Reconstructing the Human Past* (Oct 2024)
• 3rd Place, Best Oral Presentation — *6th Mexico Population Genomics Meeting, Cuernavaca* (Oct 2023)
• Young Investigator Travel Award — *SMBE, Ferrara, Italy* (July 2023)
• Elisa Acuña Scholarship — *paid internship, University of Wisconsin–Madison* (April 2023)
• CONACYT Scholarship — *stipend covering MSc and PhD studies, CINVESTAV* (Aug 2020)

Undergraduate

• Stowers Summer Scholar — *Stowers Institute, Kansas City* (Summer 2015)
• Colgate Academic Excellence — *ENES UNAM, support for research involvement* (2016)

Presentations

Talks

• Genomic history of Mexico reveals bidirectional long-distance connectivity across the Americas — *EMBO | EMBL Symposium: Reconstructing the Human Past: Using Ancient and Modern Genomics, EMBL Heidelberg, Germany* (Sept 2026)

• Inferring fine-scale demography of Indigenous lineages from 500 whole genomes across Mexico — *SMBE24, Puerto Vallarta, Mexico* (July 2024)
• Demographic modeling of admixed Latin American populations from whole genomes — *6th Mexico Population Genomics Meeting, Cuernavaca-Morelos* (Oct 2023)
• Demographic modeling of admixed Latin American populations from whole genomes — *SMBE23, Ferrara, Italy* (July 2023)

Posters

• Inferring fine-scale demography of Indigenous lineages from 500 whole genomes across Mexico — *Reconstructing the Human Past, EMBL Heidelberg* (2024)
• Deciphering mRNA decay programs during embryogenesis — *Biological Data Science Meeting, Cold Spring Harbor Laboratory* (2018)
• A pipeline for peak calling and motif discovery in ChIP-nexus data — *XVI National Congress of Plant Biochemistry, Querétaro* (2016)

Workshops & courses

• Genome bioinformatics: from short- to long-read sequencing — *EMBL-EBI, Hinxton, UK* (Nov 2025)
• Population genomics: background and tools — *EMBO courses (virtual)* (2022)
• NCBI Hackathon — *Cold Spring Harbor Laboratory, New York* (2018)
• Statistical Data Analysis for Genome Scale Biology — *EMBO, Bressanone, Italy* (2017)
• Bioinformatics Summer School — *Institute of Mathematics UNAM, Querétaro* (2015)

Teaching & Mentorship

• Research mentoring of one MSc student and one undergraduate student — Moreno-Estrada lab, CINVESTAV

Professional Service

• Manuscript reviewer — *Genome Biology and Evolution (GBE)*

References

Andrés Moreno-Estrada
CINVESTAV LANGEBIO — PhD advisor andres.moreno@cinvestav.mx

Aaron Ragsdale
University of Wisconsin–Madison — PhD co-advisor apragsdale@wisc.edu

Ariel Bazzini
Stowers Institute for Medical Research — Research supervisor arb@stowers.org